

The Two-Tape Matching Method

1 The Two-Tape Matching Method

Consider two one-dimensional tapes of identical length L .

1.1 Tape A: Buyers

For each country c , let

$$A_c = \{b_1, b_2, \dots, b_N\}$$

be the ordered set of buyers belonging to that country.

The tape is divided into N equal segments, one for each buyer. If the tape is normalized to unit length,

$$L = 1,$$

the starting point of buyer i is

$$x_i = \frac{i}{N}, \quad i = 0, \dots, N-1.$$

Each buyer therefore occupies an interval of length

$$\Delta_A = \frac{1}{N}.$$

The buyer stores only its own identifier, `buyer.uid`.

1.2 Tape B: Suppliers

For each sector s , let

$$B_s = \{s_1, s_2, \dots, s_M\}$$

be the suppliers producing intermediate goods in that sector.

Each supplier is assigned a positive weight

$$v_j = \text{targetAddedValue}_j.$$

To favour domestic suppliers, the weight is modified according to the buyer's country:

$$w_j = \begin{cases} k v_j, & \text{if } \text{country}_j = c, \\ v_j, & \text{otherwise,} \end{cases}$$

where

$$k > 1$$

is the domestic-preference factor.

The suppliers are randomly permuted once:

$$\pi : \{1, \dots, M\} \longrightarrow \{1, \dots, M\},$$

producing the shuffled tape

$$B_s^\pi.$$

The cumulative weights are then computed:

$$C_j = \sum_{h=1}^j w_{\pi(h)},$$

with

$$W = C_M = \sum_{j=1}^M w_j.$$

Supplier $\pi(j)$ occupies the interval

$$[C_{j-1}, C_j), \quad C_0 = 0,$$

whose length is

$$C_j - C_{j-1} = w_{\pi(j)}.$$

Thus, suppliers with larger weights occupy longer portions of the tape.

2 Matching the Two Tapes

The buyer tape is rescaled to the same total length W as the supplier tape. The position of buyer i becomes

$$p_i = x_i W.$$

The assigned supplier is the unique supplier at position j satisfying

$$C_{j-1} \leq p_i < C_j.$$

Geometrically, the buyer is projected vertically onto the supplier tape. The supplier interval containing that point determines the buyer's initial supplier.

No independent random draw is performed for each buyer.

3 Efficient Interval Search with `numpy.searchsorted`

The cumulative vector

$$(C_1, C_2, \dots, C_M)$$

is non-decreasing and, when all weights are positive, strictly increasing.

Given the buyer positions

$$(p_1, p_2, \dots, p_N),$$

NumPy performs the interval searches with

```
positions = np.searchsorted(
    cumulative_weights,
    buyer_points,
    side="right",
)
```

For each buyer position p_i , `searchsorted` returns the smallest index j such that

$$C_j > p_i.$$

This is precisely the index of the supplier interval containing the buyer.

The computational cost is

$$O(M)$$

for constructing the cumulative weights and approximately

$$O(N \log M)$$

for locating all buyer positions by binary search.

The searches are executed inside NumPy's compiled implementation, avoiding a Python loop over all suppliers for each buyer.

3.1 Why `side="right"`?

Supplier intervals are defined as

$$[C_{j-1}, C_j).$$

If a buyer lies exactly on a boundary,

$$p_i = C_j,$$

it must be assigned to the following interval. The option `side="right"` implements this convention.

It also skips zero-length intervals generated by suppliers whose weight is zero.

4 Systematic Placement

The recommended option is systematic placement.

Instead of using the exact beginning of every buyer segment,

$$x_i = \frac{i}{N},$$

all positions are translated by the same random offset

$$u \sim U(0, 1),$$

giving

$$x_i = \frac{i + u}{N}.$$

After rescaling to tape length W ,

$$p_i = \frac{i + u}{N}W.$$

Geometrically, the entire buyer tape is shifted by a random distance smaller than one buyer segment before being projected onto the supplier tape.

Only one random number is generated for the whole country-sector matching.

5 Alternative Placement Schemes

5.1 Starts: deterministic placement

The starting point of each buyer segment is used:

$$x_i = \frac{i}{N}.$$

Once the supplier tape has been shuffled, the matching is entirely deterministic. No additional random numbers are generated.

5.2 Systematic placement

A single common random offset is used:

$$x_i = \frac{i + u}{N}, \quad u \sim U(0, 1).$$

Its main properties are:

- only one random number is required for the entire tape;
- buyer positions remain perfectly equally spaced;
- supplier weights are represented very accurately;
- sampling variability is low;
- the geometric interpretation of the two aligned tapes is preserved.

5.3 Stratified placement

Each buyer receives an independent random position inside its own segment:

$$x_i = \frac{i + u_i}{N}, \quad u_i \sim U(0, 1).$$

Its main properties are:

- one random number is generated for each buyer;
- every buyer remains inside its own segment;
- buyer positions are no longer equally spaced;
- stochastic variability is higher than under systematic placement.

6 Comparison

Method	Random numbers	Buyer spacing	Variability
Starts	0	perfectly regular	none after shuffling
Systematic	1	perfectly regular	low
Stratified	N	irregular	higher

Table 1: Comparison of buyer-placement methods.

The systematic method provides the best compromise between computational efficiency, proportional representation of supplier weights, and the geometric interpretation of the two aligned tapes.